

# A Systematic Review of AI-Driven E-Governance and Grievance Redressal Systems

## ABSTRACT

Digital governance has become a crucial component of public service delivery, enabling citizens to interact with government institutions through online platforms. Among these, grievance redressal and civic issue reporting systems play a vital role in identifying, tracking, and resolving public issues efficiently. However, many existing e-governance platforms remain fragmented, rule-based, and limited in their ability to handle large-scale citizen data, multimedia inputs, multilingual interactions, and real time responsiveness.

This paper presents a systematic review of AI driven e-governance and grievance redressal systems, focusing on recent advancements in artificial intelligence, machine learning, natural language processing, computer vision, and conversational agents. The reviewed studies are categorized based on system functionality, including chatbot based governance platforms, mobile civic issue reporting applications, image based issue detection systems, multilingual service delivery models, and intelligent decision support mechanisms for administrators.

Through comparative analysis, this survey highlights key limitations in existing systems such as lack of end to end integration, insufficient support for regional languages, limited multimodal interaction, and weak coordination between citizens and government officials. The paper further identifies open research challenges and future directions for building intelligent, inclusive, and scalable civic engagement platforms. This review aims to serve as a foundational reference for researchers and practitioners working on next-generation smart governance and digital grievance redressal solutions.

Keywords:

E-Governance, Grievance Redressal Systems, Artificial Intelligence, Civic Issue Reporting, Smart Governance, Multilingual Chatbots, Digital Public Services

## 1. INTRODUCTION

Governments across the world are increasingly adopting digital platforms to improve the efficiency, transparency, and accessibility of public services. E-governance systems enable citizens to interact with government institutions through online portals, mobile applications, and automated service platforms, reducing dependency on manual processes and physical offices. One of the most critical components of e-governance is grievance redressal, where citizens report civic issues such as infrastructure problems, public service failures, and administrative complaints that require timely government intervention [1], [10].

Despite the widespread adoption of digital grievance platforms, many existing systems continue to face significant challenges. Traditional web-based complaint portals are often

static, form-driven, and difficult to navigate, especially for citizens with limited digital literacy. Delays in issue classification, lack of real-time tracking, limited transparency in resolution workflows, and poor coordination between citizens and government officials remain common problems [7],[9]. These limitations reduce citizen trust and hinder effective public service delivery, particularly in large-scale and diverse populations.

Recent advancements in artificial intelligence (AI) have introduced new opportunities to enhance e-governance and grievance redressal systems. Technologies such as machine learning, natural language processing (NLP), computer vision and conversational chatbots [6],[11],[12] have been applied to automate complaint classification, analyse citizen feedback, process image-based issue reports, and support multilingual communication. AI-driven systems can improve response time, reduce manual workload for administrators, and enable data-driven decision-making in public service management.

Although several research studies have proposed AI-based solutions for specific aspects of digital governance, the existing literature remains fragmented across multiple domains [1],[7], including chatbot-based service delivery, mobile civic issue reporting applications, image-based infrastructure monitoring, and multilingual interaction frameworks. Many studies focus on isolated technical components rather than integrated end-to-end governance platforms. As a result, there is a need for a systematic review that consolidates and categorizes existing research to provide a comprehensive understanding of current capabilities, limitations, and research trends.

In this paper, a systematic review of AI driven e-governance and grievance redressal systems is presented. The reviewed studies are classified based on system architecture, functional capabilities, and applied AI techniques, including conversational agents, computer vision-based issue detection, retrieval-augmented generation systems, and multilingual service interfaces. Furthermore, this survey provides a comparative analysis of existing approaches, identifies key research gaps, and outlines future research directions for building intelligent, inclusive, and scalable civic issue reporting and governance platforms.

## 2. REVIEW METHODOLOGY

This survey follows a systematic review approach to analyse recent research on AI-driven e-governance and grievance redressal systems. The objective of the review is to identify, categorize, and compare existing approaches that apply artificial intelligence techniques to digital governance, civic issue reporting, and public service delivery.

### **Literature Sources:**

The primary sources of literature for this survey include well established academic databases and digital libraries such as IEEE Xplore, SpringerLink, Elsevier ScienceDirect, MDPI, and Google Scholar. These sources were selected to ensure coverage of peer reviewed journal articles and conference proceedings relevant to e-governance, artificial intelligence, and smart governance systems.

**Search Strategy:**

A keyword based search strategy was employed to identify relevant studies. Search terms included combinations of keywords such as *e-governance*, *grievance redressal systems*, *civic issue reporting*, *AI-driven governance*, *chatbots for government services*, *image-based issue detection*, and *multilingual digital services*. The search focused on studies published between 2020 and 2025 to capture recent advancements in artificial intelligence and digital governance.

**Inclusion and Exclusion Criteria:**

Studies were included if they (i) focused on e-governance, grievance management, or civic issue reporting, (ii) incorporated artificial intelligence techniques such as machine learning, NLP, computer vision, or conversational agents, and (iii) were published in peer-reviewed journals or conference proceedings. Studies were excluded if they were non-technical opinion articles, lacked a clear methodological contribution, or focused solely on policy analysis without technological implementation.

**Paper Selection Overview:**

Based on the defined criteria, an initial pool of research articles was identified and screened by reviewing titles and abstracts. After removing duplicates and irrelevant studies, a final set of representative papers was selected for detailed analysis and classification in this survey. The selected studies were analysed with respect to system functionality, applied AI techniques, data modalities, and identified limitations.

### 3. CLASSIFICATION OF EXISTING WORK

To structure AI-driven e-governance research, literature is classified by functionality and technology, highlighting trends, strengths, and limitations in grievance redressal systems

#### 3.1 Chatbot-Based E-Governance Systems

Several studies focus on the use of conversational chatbots to improve citizen–government interaction by providing automated responses, service guidance, and grievance registration facilities. These systems are typically deployed on government websites, messaging platforms, or mobile applications to reduce manual workload and improve service accessibility. Existing chatbot-based governance solutions primarily rely on rule-based logic or AI-driven natural language processing models to handle predefined citizen queries related to government services, schemes, and complaint procedures. While such systems improve response time and availability, many lack deep contextual understanding, multimodal input support, and seamless integration with backend grievance resolution workflows [6],[11],[12],[7].

#### 3.2 Multilingual and Conversational Public Service Platforms

To address linguistic diversity and digital inclusion, several studies propose multilingual chatbot and conversational platforms for government services. These systems employ

transformer-based language models and language-specific embeddings to support regional and local languages[6],[13].

Research in this category highlights the importance of multilingual interfaces in improving accessibility for non-English-speaking populations, particularly in rural and semi-urban regions. However, challenges such as limited dataset availability for low-resource languages, reduced accuracy in mixed-language queries, and restricted support for voice-based interaction remain significant limitations.

### 3.3 AI-Based Civic Issue and Grievance Reporting Systems

Another major category of research focuses on intelligent grievance reporting systems that automate complaint classification, prioritization, and routing [10],[11] using machine learning techniques. These systems analyze structured and unstructured citizen inputs to assign grievances to relevant departments and track resolution progress. While AI-based grievance systems improve efficiency and transparency, many existing approaches operate as standalone modules and lack comprehensive integration with citizen-facing interfaces, real-time feedback mechanisms, and role-based administrative dashboards.

### 3.4 Image-Based Issue Detection and Infrastructure Monitoring

Recent studies explore the application of computer vision techniques for detecting civic issues and infrastructure problems from images and satellite data. These systems utilize deep learning models for tasks such as road damage detection, land-use change identification, and urban infrastructure monitoring [2],[5],[14],[15]. Image-based issue detection enhances the accuracy of problem identification and reduces manual inspection efforts. However, most solutions focus on specific use cases and do not integrate visual analysis with citizen-reported complaints, textual descriptions, or governance workflows.

### 3.5 Intelligent Document Retrieval and Decision Support Systems

Some studies propose intelligent document retrieval and decision support mechanisms for government platforms using retrieval-augmented generation (RAG) and semantic search techniques [3],[4]. Although such approaches improve information retrieval accuracy and reduce response time, retrieval errors, generation issues, operational complexities. Despite improvements in retrieval-augmented generation (RAG) systems, most existing approaches rely on chunk based document embeddings, which often fail to preserve semantic relationships between entities such as schemes, departments, and processes. This limitation leads to retrieval of contextually irrelevant information and increased hallucination in generated responses. Furthermore, traditional RAG systems do not explicitly model structured knowledge, making it difficult to support complex queries involving relationships or multi-step reasoning in governance scenarios.

## 4. COMPARATIVE ANALYSIS

To better understand the strengths and limitations of existing systems , a comparative analysis of representative studies is presented [1],[6],[7],[10] focusing on key system features, applied AI techniques, supported data modalities, and identified limitations.

| Ref. Category                                   | Core Functionality                                 | AI Techniques Used                   | Multilingual Support | Image-Based Reporting | End-to-End Integration | Key Limitations                                                |
|-------------------------------------------------|----------------------------------------------------|--------------------------------------|----------------------|-----------------------|------------------------|----------------------------------------------------------------|
| Chatbot-based governance systems                | Automated citizen interaction and service guidance | NLP, rule-based models, transformers | Limited              | No                    | Partial                | Lack of contextual understanding and workflow integration      |
| Multilingual governance platforms               | Regional language service delivery                 | Transformer-based NLP models         | Yes                  | No                    | Partial                | Low accuracy for low-resource languages, limited voice support |
| AI-based grievance reporting systems            | Complaint classification and routing               | Machine learning, text analytics     | Limited              | No                    | Partial                | Standalone operation, weak citizen feedback mechanisms         |
| Image-based issue detection systems             | Visual detection of civic issues                   | CNNs, vision transformers            | No                   | Yes                   | No                     | Narrow use cases, poor integration with text-based complaints  |
| Document retrieval and decision support systems | Policy and document access                         | RAG, semantic search                 | Limited              | Yes (document images) | No                     | Not designed for grievance lifecycle management                |

Table I: Comparison of AI-Driven E-Governance and Grievance Redressal Systems

Observational Analysis:

The comparative analysis reveals that most existing systems address specific components of digital governance in isolation. Chatbot-based systems improve accessibility but lack deep integration with grievance resolution workflows. Multilingual platforms enhance inclusivity but face challenges related to language resource availability and voice interaction accuracy. Image-based issue detection systems offer precise visual analysis but are rarely connected to citizen-facing reporting platforms. Similarly, document retrieval systems improve information

access but do not support complete grievance management. Overall, there is a clear absence of integrated, multimodal, and role-based governance platforms that unify citizen interaction, intelligent issue analysis, and administrative decision-making [7],[10],[15].

## 5. RESEARCH GAPS AND CHALLENGES

Despite significant progress in AI driven e-governance and grievance redressal systems, several research gaps and practical challenges remain unresolved. The analysis of existing literature reveals that most proposed solutions focus on isolated components of governance platforms rather than delivering fully integrated and citizen-centric systems.

One major research gap is the absence of end-to-end grievance management platforms that seamlessly integrate citizen interaction, intelligent issue analysis, administrative workflows, and resolution tracking within a unified architecture. Existing systems often operate as standalone modules such as chatbots, complaint classifiers, or image analysis tools without effective coordination across the complete grievance lifecycle [7],[10].

Another critical challenge lies in multimodal data integration. While some studies utilize text-based complaint analysis or image-based issue detection independently, very few systems support the combined processing of textual descriptions and images submitted by citizens. This limitation restricts accurate issue understanding and reduces the effectiveness of automated decision making in real world governance scenarios [2],[5],[15].

Multilingual and voice based **accessibility** remains an open research problem, particularly in linguistically diverse regions. Although several platforms support multilingual text interaction, limited availability of high-quality datasets for regional languages, mixed-language queries, and speech to text accuracy issues hinder inclusive access to digital governance services for rural and semi-literate populations [6],[13].

Further, most existing systems lack role-based coordination between citizens, field level supervisors, and administrative authorities. The absence of transparent progress tracking, real time status updates, and visual resolution evidence reduces citizen trust and accountability in grievance redressal processes [9],[10].

From an operational perspective, challenges related to scalability, data privacy, and system adaptability are often underexplored. Many studies are evaluated in controlled or small-scale environments, with limited discussion on large-scale deployment, secure data handling, and adaptability to evolving governance policies and service requirements [1],[7].

Another critical research gap lies in the lack of structured knowledge representation in existing retrieval based systems. Most current approaches rely on unstructured or semi-structured document embeddings, which fail to capture relationships between key governance entities such as schemes, departments, complaint types, and processes. This limitation reduces retrieval precision and hinders reasoning capabilities in complex query scenarios. There is a need for graph based knowledge modeling approaches that can represent entity relationships explicitly while supporting efficient retrieval mechanisms.[11],[12],[7].

These gaps highlight the need for intelligent, scalable, and inclusive e-governance platforms that integrate multimodal citizen interaction, AI driven issue analysis, role based workflows, and transparent grievance lifecycle management.

## 6. FUTURE RESEARCH DIRECTIONS

Future research in AI-driven e-governance and grievance redressal systems should focus on developing integrated, citizen-centric platforms that unify multiple functionalities within a single architecture. Rather than addressing isolated technical components, future systems should support the complete grievance lifecycle, from issue reporting to resolution verification and feedback collection.

Advancements in multimodal AI present promising opportunities for improving civic issue understanding by jointly analyzing text, images, and voice inputs. Research efforts should explore robust multimodal fusion techniques that enhance issue classification accuracy and contextual understanding in diverse real-world environments.

Enhancing multilingual and voice-enabled interaction remains a critical direction for inclusive digital governance. Future studies should focus on improving speech-to-text and natural language understanding capabilities for low-resource and mixed-language settings, enabling wider participation from rural and linguistically diverse populations.

The integration of real-time analytics and predictive intelligence can further improve decision-making in governance platforms. Predictive models may assist authorities in identifying recurring issues, prioritizing critical complaints, and optimizing resource allocation across departments.

A promising direction for future research is the development of graph-centric retrieval frameworks that integrate knowledge graphs with semantic retrieval mechanisms. In such systems, documents can be transformed into structured knowledge graphs by extracting entities and relationships. Each graph can then be summarized into a semantic description and embedded for efficient retrieval. At query time, relevant graphs can be identified using embedding similarity, followed by graph traversal to extract precise and contextually grounded information. This approach has the potential to significantly reduce hallucination, improve retrieval accuracy, and support complex reasoning in e-governance applications.

Such hybrid graph-based retrieval systems can enable dynamic knowledge integration, allowing governance platforms to adapt to continuously evolving policies and schemes without requiring frequent model retraining.

Finally, future research should address scalability, privacy, and security considerations to support large-scale deployment of intelligent governance systems. Ensuring secure data handling, compliance with regulatory frameworks, and adaptability to evolving government policies [1],[7] will be essential for the successful adoption of next-generation e-governance platforms.

## 7. CONCLUSION

This paper presented a systematic review of AI-driven e-governance and grievance redressal systems, focusing on recent research that applies artificial intelligence techniques to digital public service delivery. Existing studies were classified into key categories, including chatbot-based governance platforms, multilingual service systems, AI-driven grievance reporting solutions, image-based issue detection approaches, and intelligent document retrieval mechanisms.

Through comparative analysis, the survey highlighted that while individual components of intelligent governance have advanced significantly, most existing solutions remain fragmented and lack comprehensive end-to-end integration. Key challenges identified include limited multimodal interaction support, insufficient multilingual and voice-enabled accessibility, lack of role-based coordination, and weak transparency across grievance resolution workflows.

By identifying these research gaps and outlining future research directions, this survey provides a consolidated reference for researchers and practitioners working toward the development of intelligent, inclusive, and scalable e-governance platforms. The insights presented in this review aim to support the design of next-generation civic issue reporting and grievance redressal systems that enhance citizen engagement, operational efficiency, and trust in digital governance.

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